Stackly Finance

Website Technical Documentation

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Vanilla HTML, CSS and JavaScript | 16 pages | Flat single-folder project

Audience: developers, testers and reviewers

Prepared for the Stackly Finance project team. Contents describe the front-end demo build; figures and content are placeholders.

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1. Overview and scope

Stackly Finance is a 16-page marketing and demo website for a finance brand offering personal loans, home and auto loans, insurance and mutual funds. It is built with vanilla HTML, CSS and JavaScript only, in one flat folder, and is ready for GitHub Pages. It includes a working EMI calculator, validated forms, a role-based login (User or Admin) with two separate dashboards, and a luxury midnight-and-gold visual design.

In scope: all public pages, shared styles and script, forms, calculator, front-end authentication demo, dashboards, 404 page, responsive layout and accessibility basics. **Out of scope:** any server, database, email delivery, payment or real identity checks.

2. Technology and constraints

Item	Detail
Languages	HTML5, CSS3 (custom properties, grid, flexbox, clamp), ES5-style JavaScript in strict mode
Frameworks / build	None. No bundler, no package manager, no runtime dependencies
External resources	Google Fonts only (Cormorant Garamond for headings, Manrope for body). Everything else is local
Images	WebP (plus one PNG favicon). Every image is under 100 KB and has alt text
Browser storage	localStorage (demo accounts), sessionStorage (current session and one-time prefill)
Hosting	Any static host. Tested layout for GitHub Pages (relative links, root-level files)
Size	16 HTML files, style.css (477 lines), script.js (498 lines), 18 images (167 KB total, largest hero.webp at 14 KB)

3. Project structure

All files sit in a single folder. There are no sub-directories, which keeps links simple and works unchanged on GitHub Pages.

File(s)	Purpose
index.html, about.html, services.html, personal-loans.html, home-auto-loans.html, insurance.html, investments.html, calculator.html, pricing.html, blog.html, faq.html, contact.html	The 12 public content pages
login.html	Log in and Sign up in one page (tabs), User or Admin
user-dashboard.html, admin-dashboard.html	Role-specific dashboards (guarded)
404.html	Target for placeholder links (social icons, some footer links)
style.css	All styles: base, components, luxury theme, QA fixes, dashboard bar
script.js	All behaviour: shared header/footer, menu, calculator, validation, auth, dashboards
logo-light.webp, logo-color.webp, favicon.png	Brand logo for dark backgrounds, original colours, and tab icon
hero.webp, about.webp, avatar-*.webp, team-*.webp, blog-*.webp	Content images (each under 100 KB)
README.md	Short setup and deployment guide

4. Page inventory

Page	File	Highlights
Home	index.html	Hero with live mini EMI estimator, six service cards, animated stats, testimonials, CTA
About Us	about.html	Story, values, stats band, leadership team
Services	services.html	Six service cards, how-it-works steps
Personal Loans	personal-loans.html	Features, terms table, eligibility, application steps
Home & Auto Loans	home-auto-loans.html	Same template, two-column terms table
Insurance	insurance.html	Health, term, motor plans table
Investments & Mutual Funds	investments.html	Fund type, horizon and risk table with risk disclaimer
EMI Calculator	calculator.html	Sliders and number boxes, live result, donut chart, yearly schedule
Pricing	pricing.html	Three plans with a featured plan
Insights (Blog)	blog.html	Six articles, topic filter chips, expandable full text
FAQ	faq.html	Accessible one-open-at-a-time accordion
Contact	contact.html	Validated form with per-field errors and summary
Log in / Sign up	login.html	Role choice, password policy checklist, demo buttons, no footer
User dashboard	user-dashboard.html	Loan progress, EMIs, investments, profile, activity
Admin dashboard	admin-dashboard.html	Stats, monthly chart, applications table with approve/reject and search, users, tickets
Not found	404.html	Message with Home and Contact buttons, no footer

5. Architecture

Shared shell. Each page contains empty `<header id="site-header">` and `<footer id="site-footer">` elements. `script.js` fills them from one source, so the menu, logo and footer are edited in one place. The page sets `<body data-page="...">`; the script uses it to mark the active navigation link (product pages use the key "services"). Pages that need a different shell omit the elements: the login and 404 pages have no footer, and the dashboards have their own top bar and no footer.

Progressive behaviour. A tiny inline script adds the class "js" to the root element so reveal animations only apply when JavaScript runs. A single IntersectionObserver handles scroll reveals and number counters, with a short stagger between items.

Icons. An inline SVG sprite is injected once per page and referenced with `<use href="#i-name">`. Social icons use the same sprite.

Static pages. The delivered pages are plain static HTML files that can be edited by hand. No server-side code runs.

6. Design system

Theme: luxury midnight and gold, with a white and platinum content background.

Token	Value	Use
Ink / navy	#0c1220	Headings, header, dark sections
Midnight blue	#1b2742	Gradients, secondary dark

Token	Value	Use
Gold (text and buttons on light)	#8c6a1c	Links, icons, accents with accessible contrast
Gold gradient	#f3dd9f / #c9a24b / #a07a22	Primary buttons, progress bar, active states
Champagne	#f0d994	Accents on dark backgrounds
Platinum	#f2f1ed	Alternate sections, table headers
Heading font	Cormorant Garamond 500-700	Headings, numbers (lining figures enabled)
Body font	Manrope 400-700	Text, forms, buttons

- **Components:** cards, icon boxes, pill buttons (primary gold, ghost, light), tables, pricing plans, accordion, chips, tabs, notices, progress and bar charts, status badges.
- **Motion:** staggered hero entrance, floating EMI card, scroll-progress bar, animated nav underline, card lift and icon rotation on hover, button shimmer, image zoom on blog cards, counters, filling progress and chart bars.
- **Reduced motion:** a single prefers-reduced-motion rule disables transitions and animations.
- **Hidden elements:** a global rule keeps the HTML hidden attribute working even when a component sets its own display value.

7. Responsive behaviour

Breakpoint	Changes
1180 px and below	Navigation becomes a hamburger menu; four-column grids become two; footer becomes two columns with the brand and Navigate blocks full width
900 px and below	Hero stacks; the EMI card sits under the image; split layouts and the calculator become one column; three-column grids become two
600 px and below	Three-column grids become one; steps become one column; forms become single column; dashboard title hides in the top bar

Layouts were checked for horizontal scrolling at 360, 375, 414, 768, 1024, 1180, 1280, 1366, 1440 and 1920 px.

8. JavaScript modules

`script.js` is a set of self-contained blocks inside two immediately-invoked functions. A small public object, `window.Stackly`, exposes the EMI formula, rupee formatter and slider-fill helper.

Block	Responsibility
Header and footer	Builds navigation, active link, logo, social icons and footer columns; shows "Dashboard" in the header when a session exists
Mobile menu	Toggle with aria-expanded, closes on link click, Escape and outside click
Scroll effects	Header shadow, scroll-progress bar, reveal and counters
EMI helpers	emi(P, annualRate, months) uses the standard reducing-balance formula; inr() formats in the Indian grouping
Mini estimator	Home page sliders with live monthly EMI at a fixed 10.5 percent
Calculator	Range and number inputs stay in sync, out-of-range values show an error and are clamped on blur; renders totals, donut and a yearly table
Form engine (initForm)	Field-level validation on blur and input, summary message above the submit button, focus on first error
Password module	Seven-rule checklist, honest strength label, show/hide toggle

Block	Responsibility
Auth module	Sign up, log in, session, demo buttons, admin access code
Dashboard guard	Shows the dashboard only to the matching role, otherwise shows a clear message with a button
Admin tools	Approve and reject applications with a live status message; search filter
Accordion and blog filter	One FAQ item open at a time; topic chips filter posts

9. Forms and validation

Every form uses novalidate and the shared validator. Each field has a data-label (for readable messages), an optional data-rule, a visible label and an error paragraph linked with aria-describedby. Errors show on blur, update while typing once touched, and all invalid fields are listed on submit with focus moved to the first one.

Field	Rule
Full name	Letters, spaces, apostrophe, dot, hyphen; 2 to 60 characters; must start with a letter
Email	Strict pattern: no consecutive dots, local part up to 64 characters, valid domain and 2+ letter ending
Mobile number	Exactly 10 digits starting 6 to 9. Spaces, dashes, letters and a +91 prefix are removed as you type or paste
Message (contact)	Required, at least 10 characters
Topic / account type	Must be chosen
Consent / terms	Checkbox must be ticked
Confirm password	Must equal the password and revalidates when the password changes
Admin access code	Shown only for Admin sign-up; must match the code (case-insensitive)

Password policy (sign up)

- At least 10 characters, with an uppercase letter, a lowercase letter, a number and a symbol.
- No spaces and no character repeated three times in a row.
- No common words (for example password, qwerty, welcome, admin, stackly).
- No part of the user name (4+ letters) and no email name inside the password.
- The checklist shows each rule live. The strength label reads Good or Strong only when every rule passes (Strong at 14+ characters), so the checklist always matches what the form accepts.

10. Authentication, roles and dashboards

Important: this is a front-end demo. There is no server. It is suitable for demonstrating flows, not for protecting data.

Topic	Behaviour
Accounts	Stored in localStorage key stackly_users: name, email, role, SHA-256 hash of "stackly:email:password" (a simple fallback hash is used if the browser has no crypto.subtle)
Session	sessionStorage key stackly_session: name, email, role. Cleared on log out and when the tab closes
Roles	User and Admin, chosen on both the Log in and Sign up forms
Demo accounts	user@stackly.demo and admin@stackly.demo, built into the script (not stored)

Topic	Behaviour
Admin sign-up	Requires an access code held in script.js, so it is visible to anyone who reads the source

Flow

- **Sign up:** valid form saves the account, then redirects to `login.html?created=1#login`. The login form shows "Account created successfully", pre-fills the email and account type, and focuses the password. The user is not logged in automatically.
- **Log in:** the account type must match the account. Success opens the matching dashboard: User to user-dashboard.html, Admin to admin-dashboard.html. Errors say whether the email was not found, the password was wrong, or the account type was wrong.
- **Dashboard access:** logged out shows "Please log in to continue" with a Go to login button. The wrong role shows a message and a button to the correct dashboard.
- **Dashboard top bar:** no site navigation. Only the logo (not a link), the page title, a Back button and a Log out button. Both buttons lead to login.html; Log out also clears the session.
- **Login page with a session:** no automatic redirect; a note offers "Open your dashboard".

Dashboard	Content
User	Welcome header, four stat cards (outstanding loan, next EMI, investments, credit score), loan progress bar, upcoming EMI table, investments table, profile, quick actions, recent activity
Admin	Welcome header, four stat cards, applications table with Approve / Reject and search, recent users, monthly applications bar chart, open support tickets

11. Accessibility and SEO

- Semantic landmarks (header, nav, main, footer, section, article, figure, table with captions), one h1 per page, skip link, visible focus ring.
- All images have alt text; decorative icons are aria-hidden; social links have aria-labels.
- Forms: labels linked to inputs, errors linked with aria-describedby, aria-invalid, aria-live regions, role=alert for banners, show/hide password button with aria-pressed.
- Tabs use role=tablist/tab/tabpanel; accordion uses buttons with aria-expanded and aria-controls.
- Colours chosen for contrast: gold on light backgrounds uses the darker #8c6a1c; dark sections use champagne text.
- Each page has a unique title, meta description, Open Graph tags and theme colour.

12. Performance and assets

- No JavaScript or CSS frameworks; two local files plus Google Fonts.
- Images are WebP, sized to their display use, lazy-loaded below the fold, with width and height set to avoid layout shift. The hero image is fetched with high priority.
- Total image weight is about 167 KB for all 18 images.
- Animations use transform and opacity only.

Image	Size (KB)
hero.webp	14.5
logo-color.webp	13.8
logo-light.webp	10.5

Image	Size (KB)
blog-emi.webp	10.3
blog-fund.webp	9.5
about.webp	9.3
team-1.webp	9.0
team-4.webp	8.9
team-2.webp	8.9
team-3.webp	8.9
blog-insurance.webp	8.7
favicon.png	8.3
blog-home.webp	8.1
blog-checks.webp	7.8
avatar-karthik.webp	7.7
avatar-meena.webp	7.7
blog-sip.webp	7.6
avatar-lakshmi.webp	7.4

13. Testing and QA results

An automated Playwright suite (Chromium) ran **113 checks with 0 failures** on the final build. Firefox and Safari engines were not available in the test environment, so those browsers still need manual testing.

Area	What was verified
Sign up and login	Weak and repeated-character passwords rejected with visible errors; sign-up redirects to login with a success message and prefilled email; login to each dashboard; wrong role, wrong password and unknown email messages; demo buttons
Dashboards	No site navigation or footer; top bar has only Back and Log out, both to login; Back does not bounce; Log out clears the session; guard messages for logged-out and wrong-role visits; approve, reject and search work
Forms	Mobile number cannot contain spaces; +91 and dashes are cleaned; number starting with 5 rejected; summary and per-field errors
Layout	Login page centred with no footer at 7 widths; 404 has no footer; cards aligned in rows on Home, Services, Blog, Pricing and product pages; footer columns end together; header on one line from 1181 to 1920 px; hamburger at 1180 and below
Resilience	No horizontal scroll on any page at 10 widths, including both dashboards with a session; no JavaScript errors
Other	Blog filters, calculator values, image sizes, link targets

Suggested manual checks

- Real phones and tablets, plus Firefox and Safari.
- Keyboard-only navigation of the menu, forms, tabs and accordion.
- Screen reader pass on the login and dashboard pages.

- Reload behaviour on a deployed GitHub Pages URL (accounts are per browser and per address).

14. Bug and change log

A. Tester report (8 items) and related fixes

#	Issue	Cause	Fix
1	Create account looked unclickable	Password rejected for 3 repeated characters while the checklist and meter showed all clear; error was small and far from the button	Seven-rule checklist, honest strength label, red error text, summary above the button
2	Login failed with a vague message	No account existed for that email; message did not say so	Specific messages for unknown email, wrong password, wrong account type; Demo user and Demo admin buttons
3	Login page not aligned	Hero text left, card centred, uneven tabs	Centred card on a dark full-height page, equal-width tabs
4	No footer wanted on login and sign up	Footer rendered on all pages	Footer removed from login.html
5	Mobile number accepted spaces	Validator stripped spaces before testing	Digits-only input filter and strict rule on Contact and Sign up
6	Service and blog cards not aligned	Links floated at different heights	Equal-height flex cards with links pinned to the bottom
7	Footer columns uneven	Navigate had 8 stacked links against 4-5 elsewhere	Navigate in two columns; all columns four rows
8	Dashboards would not open	Chain from items 1 and 2; logged-out visits redirected silently	Clear on-page message and button instead of silent redirect

B. Extra bugs found during retesting

Issue	Fix
Header links wrapped to two lines between 1024 and 1180 px	Hamburger breakpoint raised to 1180 px
Error text inside cards rendered grey and lost spacing	Higher-specificity colour and spacing rules
New flex layouts overrode the HTML hidden attribute (showed the login gate on working dashboards; would have broken the blog filter)	Global hidden rule with !important
Checkbox could be enlarged by input padding	Explicit 18 px checkbox size

C. Latest change requests

Request	Implementation
After creating an account, go to the login page	Redirect to login.html?created=1#login with a success message and prefilled email and role; no automatic login
Remove the footer from the 404 page	404.html built without a footer
Dashboard pages: no navigation; only Back and Log out, both to login	Dedicated top bar (logo, title, Back, Log out); site header and footer removed from both dashboards; page-level extra buttons removed; login page no longer auto-redirects when a session exists, and shows a note instead

15. Deployment (GitHub Pages)

- Create a repository and place every file at its root (keep the flat structure).

- Commit and push to the main branch.
- Open Settings, then Pages. Set Source to "Deploy from a branch", branch main, folder / (root), and Save.
- Wait a minute or two, then open https://<username>.github.io/<repository>/.
- GitHub Pages serves 404.html automatically for unknown addresses.
- Accounts created in the demo exist only in the visitor browser on that address. After redeploying, ask testers to hard-refresh (Ctrl+F5) to avoid cached files.

16. Maintenance guide

To change	Do this
Menu items	Edit the NAV list at the top of script.js (file, label, data-page key)
Footer links	Edit SERVICES, COMPANY and the footer block in script.js. Placeholder links point to 404.html; replace with real pages when ready
Logo	Replace logo-light.webp (dark backgrounds) and logo-color.webp (light backgrounds); a high-resolution SVG or PNG gives the sharpest result. Update favicon.png
Colours and fonts	Change the custom properties at the top of style.css, then the "Luxury theme" block
Contact details	Footer block in script.js and the Visit or call card on contact.html
Rates, fees, text	Edit the relevant HTML page directly
Demo credentials and admin code	DEMO list and ADMIN_CODE in script.js (visible to anyone; replace with a server-side check before real use)
Add a page	Copy an existing page, change title, meta and data-page, add it to NAV

17. Known limitations and recommendations

- **No backend.** Authentication, accounts and the admin code are client-side only. A real launch needs a server for authentication, authorisation, password handling, the contact form and persistence.
- **Placeholder content.** Phone, email, address, statistics, testimonials, team, rates, fees and dashboard figures are illustrative.
- **Logo quality.** The logo was extracted from a low-resolution image and may look soft on large displays. Supply an SVG or high-resolution PNG.
- **Placeholder links.** Social icons and several footer links intentionally open 404.html.
- **Browsers.** Automated tests ran on Chromium only.
- **Fonts.** Google Fonts needs an internet connection; local fallbacks are defined.
- **Recommendations.** Add privacy and terms pages, connect the contact form to an email service, add analytics and cookie consent if required, and run an independent accessibility audit.

Appendix: demo credentials

For testing only. These are public and must not be used on a live system.

Item	Value
User login	user@stackly.demo
Admin login	admin@stackly.demo
Password (both)	Stackly@2026!
Admin sign-up access code	STACKLY-ADMIN-2026